

**(주)동인엔시스**

COMPANY PROFILE 2026

# Controlling equipment, connecting data.

Smart manufacturing solutions since 1991 · Intelligent Control Panel

[donginensis.com](http://donginensis.com)

# After 35 years of building control panels, we are turning them into data infrastructure.

Since 1991, DONG-IN ENSIS has designed and built control panels for industrial and marine sites. On that foundation, we now make Intelligent Control Panels that collect and connect equipment data while controlling it.



Intelligent Control Panels on a factory floor · illustration

## 35<sup>+</sup>

### Years

Founded in 1991 as a supplier of industrial control equipment

## 750<sup>+</sup>

### Projects

Marine, energy and plant control system projects

## 15<sup>+</sup>

### Industries

Shipbuilding, hydrogen, energy, manufacturing, data centers

## 40<sup>+</sup>

### CE·UL

CE·UL certified solutions

# Industrial DX and AX start with field data.



Thank you for visiting DONG-IN ENSIS.

We began in 1991 as a supplier of industrial control equipment and grew into a company that designs and builds control systems for ships, energy facilities and plants. Over more than 750 projects, one thing became clear:

most equipment is already controlled by PLCs and drives, but the data they generate never leaves the control panel.

AI and smart factories need field data first. Our Intelligent Control Panel collects data where the equipment is controlled and delivers it to the cloud or customer servers in a standard format. For manufacturers, it is the first step toward managing equipment with data. For IT companies, it is a reliable source of field data.

**We will keep adding data and intelligence to the control technology we have built in the field for 35 years.**

CEO, DONG-IN ENSIS CO., LTD. **Min-gi Baek**

# Company overview

|              |                                                                                                                                                                                                                                                                             |                                                                        |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Company      | DONG-IN ENSIS CO., LTD.<br>(주)동인엔시스 (formerly Dongin Gijeon)                                                                                                                                                                                                                | Slogan Controlling equipment, connecting data.<br>설비를 제어하고, 데이터를 연결하다. |
| CEO          | Min-gi Baek                                                                                                                                                                                                                                                                 |                                                                        |
| Founded      | October 1991<br>Incorporated July 1997 · renamed 2023                                                                                                                                                                                                                       |                                                                        |
| Business     | Intelligent Control Panels · industrial and marine control systems<br>System integration · supply of global power and control products                                                                                                                                      |                                                                        |
| Sites        | Head office, Busanjin-gu, Busan, Korea<br>Eco-Technology Research Center (NS Tower) · Yangsan Factories 1 & 2 · Gijang Factory                                                                                                                                              |                                                                        |
| Management   | ISO 9001 · ISO 14001 · ISO 45001                                                                                                                                                                                                                                            |                                                                        |
| Designations | Venture Company · INNOBIZ · MAINBIZ · Corporate R&D Center · Smart Factory Supplier<br>Hydrogen Specialized Company · Innovation Premier 1000 (2026) · National Innovative Company 1000<br>Busan Strategic Industry Leading Company · Legend 50+ · Busan Service Strong SME |                                                                        |
| Partners     | Schneider Electric official SI partner<br>Supply partner for ABB · Fuji Electric · Eaton · Danfoss                                                                                                                                                                          |                                                                        |

VISION

Accelerating **DX and AX** in manufacturing and maritime industries with field equipment data.

MISSION

Creating and connecting **reliable data** from the control panels that run the equipment.



Intelligent Control Panel at a hydrogen facility · illustration

# From components to systems, from systems to data



Why We Can

# Three foundations behind the Intelligent Control Panel

Data infrastructure is not built with software alone. It starts with hands-on experience controlling real equipment.

## 01 · MANUFACTURING

### We design and build in-house

From panel design to build, wiring, PLC programming, factory testing and on-site commissioning.

- Yangsan Factories 1 & 2 and Gijang Factory
- Dedicated FAT test line
- Design and build to CE·UL standards

## 02 · FIELD EXPERIENCE

### We control demanding sites

Control systems built and verified at hydrogen, shipbuilding and plant sites with strict safety requirements.

- Liquid hydrogen station control nationwide
- FGSS test facility control for a major shipyard
- Hybrid electric propulsion for green ships

## 03 · R&D

### We research data and predictive control

Our Eco-Technology Research Center leads national and regional R&D and industry-academia work.

- 5 registered patents on LH2 station control
- National R&D on hydrogen-powered vessels
- Joint research with Korea Maritime & Ocean Univ.

Global brand partners

Schneider Electric official SI partner

ABB

Fuji Electric

Eaton

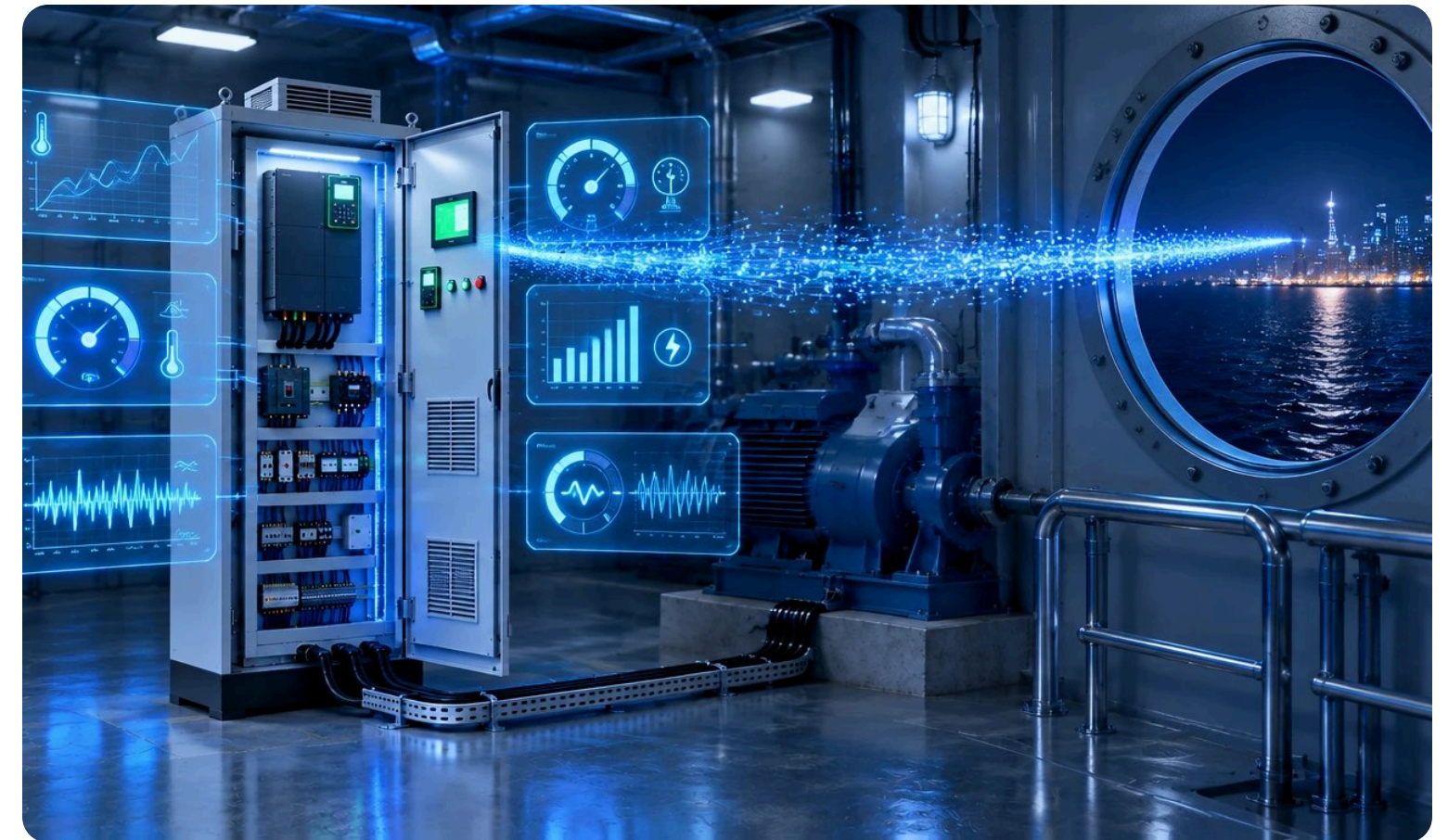
Danfoss

Intelligent Control Panel

# From equipment control to data connectivity, one Intelligent Control Panel.

It controls equipment with a PLC while collecting operating data, and connects securely to the cloud or customer servers without extra data collection devices.

Motor winding temp. U·V·W   Bearing temp.   Speed   Voltage · current · energy  
Vibration   Ambient temp. · humidity   Run status



Intelligent Control Panel collecting and connecting equipment data · illustration

## 01

### Real-time PLC control & data collection

Control logic and interlocks stay with the field PLC.

## 02

### MQTT / TLS transmission

The PLC sends operating data over MQTT, protected by TLS.

## 03

### DataHub standard time series

Equipment data is organized in a common structure and stored over time.

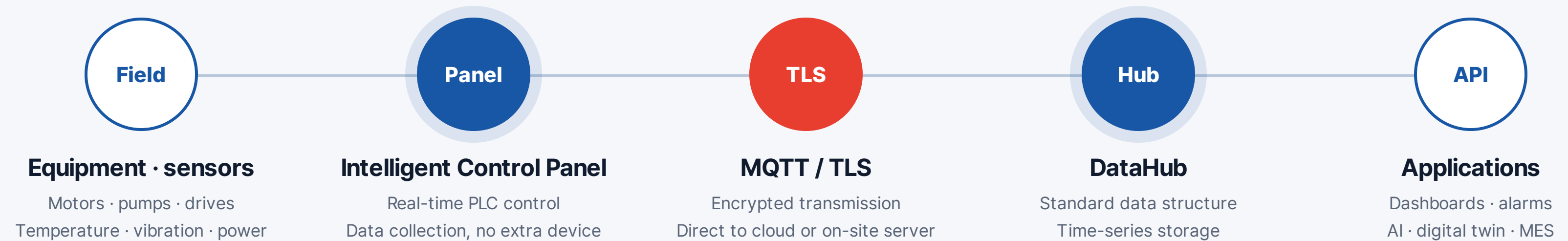
## 04

### REST API integration

REST API and WebSocket for MES, monitoring and analytics systems.

How It Works

# From field equipment to AI, seamlessly connected



Cloud

## Connect to the DONG-IN ENSIS cloud

- Fast deployment with no server to set up
- Check anytime on web and mobile
- Manage multiple plants and sites on one screen

On-premise

## Install on customer servers and networks

- DataHub and dashboard run in-house, without internet access
- Use the customer's own server or one supplied by us
- For sites whose security policy rules out external cloud

Both options use the same data structure and API. Maintenance for on-premise systems is agreed in line with the customer's security policy.

# Equipment safety and data security, with no extra collection devices

## 01 Control stays in the field

Interlocks and final control remain with the field PLC. Servers never control equipment directly.

## 02 Encrypted transmission

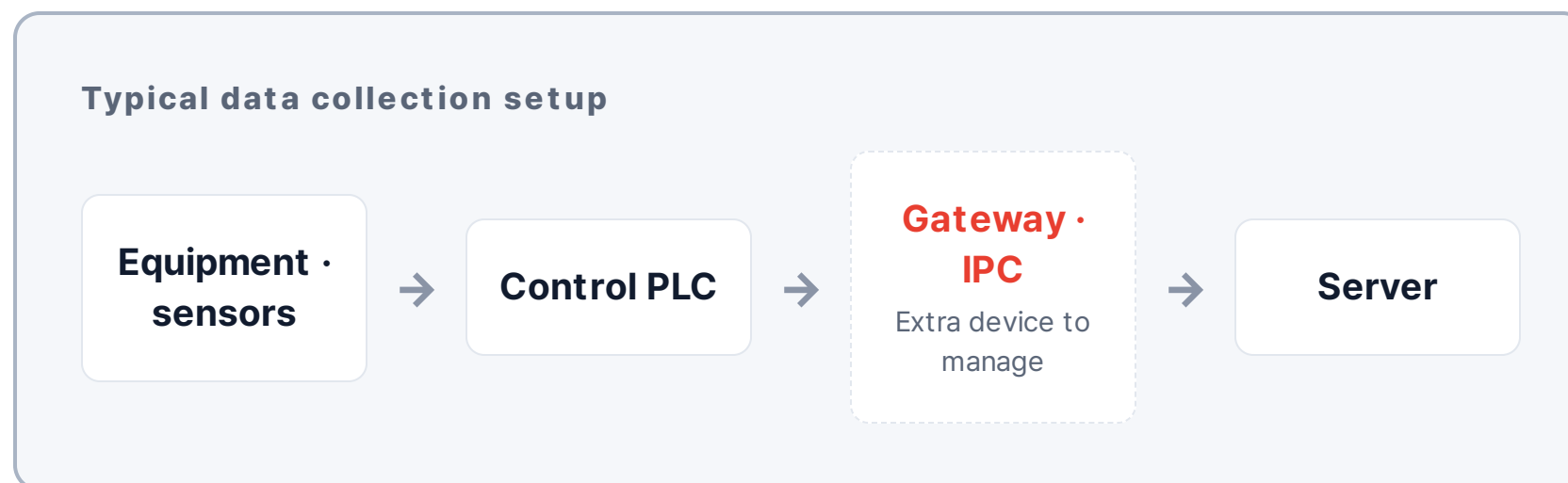
MQTT traffic between the site and the cloud (or customer server) is encrypted with TLS.

## 03 Runs without external networks

On-premise systems run without internet access; cloud systems can connect over LTE, separate from the plant network.

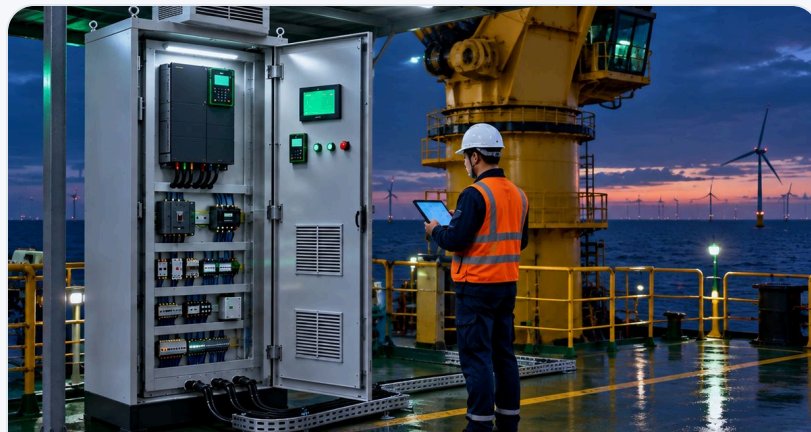
## 04 Access and data use

Accounts, permissions, scope of data use and retention periods are set in each project contract.



\* Network and security configuration is agreed according to site conditions and the customer's security policy.

# From the sea to data centers, wherever motors run



## MARINE · OFFSHORE

### Shipbuilding · marine

Control proven on a moving sea

- LNG carrier engine rooms
- Ballast water treatment
- Wind installation vessels · port cranes



## HYDROGEN · ENERGY

### Hydrogen · energy

Hydrogen control backed by patents

- Liquid hydrogen stations
- Hydrogen-powered vessels
- 5 patents · Hydrogen Specialized Co.



## MANUFACTURING

### Smart factory

Start a smart factory with a panel swap

- Steel blowers · air compressors
- Water pumping stations
- Production lines · food & pharma



## DATA CENTER

### Data centers

Nonstop cooling, lower power

- Chillers · cooling water pumps
- Cooling tower fans · liquid cooling CDU
- DCIM · BMS integration

Water booster pump  
station  
160kW × 3

Steel mill blower  
250kW

Air compressor room  
90kW × 2

Liquid hydrogen  
station  
30kW

LNG carrier engine  
room  
75kW × 2

Ballast water  
treatment  
55kW

Wind installation  
vessel crane  
200kW

Port container crane  
300kW

\* Application scenarios are examples. Images are illustrations.

# Green ships and hydrogen facilities, proven at the most demanding sites



## HYDROGEN VESSEL

### Hydrogen-powered vessels

- Ministry of Oceans and Fisheries project on small hydrogen-powered vessel technology and demonstration
- Integrated control of fuel cells, propulsion motors and hydrogen tanks
- Predictive power control and output distribution



## HYBRID PROPULSION

### Hybrid electric vessels

- Hybrid propulsion and integrated control for the Chungnam Hospital Ship and Gyeongnam Cheongjeong (2023)
- Diesel-electric integrated control for a pilot hybrid fishing vessel (2026)
- National R&D on hybrid fishing vessel equipment (2021)



## LIQUID HYDROGEN

### Liquid hydrogen stations

- Control systems for a liquid hydrogen station project led by a major Korean company
- Field-proven at commercial stations
- 5 control system patents · Hydrogen Specialized Company

\* Images are illustrations.

# Hybrid propulsion for green vessels, delivered with integrated control



## Chungnam Hospital Ship · G/T 330 t

- Diesel engine + electric motor hybrid propulsion
- Propulsion system · integrated control & monitoring
- Commissioned Aug 2023 · green-ship commendation



## Environmental cleanup ship · G/T 120 t

- Diesel engine + electric motor hybrid propulsion
- Propulsion system · integrated control & monitoring
- Commissioned Apr 2023



## Harbor cleaning ship · G/T 280 t

- Diesel & LNG engine hybrid propulsion
- Fuel gas supply system (FGSS) control
- Gas safety control on an LNG-fuelled vessel

※ Hybrid demonstration fishing vessel: diesel-electric integrated control (2026) · National R&D: electric hybrid fishing vessels (2021-25), safety-based small hydrogen vessels (2022-26)

Major Reference · LNG Fuel Gas Supply System

# FGSS control systems, proven on LNG-fuelled ships

| Owner · Shipyard                       | Vessel · Facility                                                     | Scope of supply                                                      |
|----------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------|
| Jiangsu New Times Shipbuilding (NTS)   | EPS-NTS 210,000 / 209,000 DWT bulk carriers, 13 ships (Liberia · ABS) | FGSS glycol water pump control · hydraulic power supply system       |
| Namura Shipbuilding                    | 95,000 DWT bulk carrier (S496 · Liberia · NK)                         | FGSS glycol water pump control system                                |
| GSI (Guangzhou Shipyard International) | HMM 8,600 CEU DF pure car truck carriers, 3 ships (Panama · KR)       | FGSS glycol pump · L.O pump & oil heater control                     |
| GSI (Guangzhou Shipyard International) | EPS/Chandris 111K DWT LNG DF crude/product tankers (Taiwan · KR)      | BOG compressor control · glycol pump · L.O pump & oil heater control |
| —                                      | 1.2 MW anchor handling ship (EK011 · Taiwan · KR)                     | FGSS glycol pump · L.O pump & oil heater control                     |
| Samsung Heavy Industries               | FGSS system test facility                                             | G/W (glycol water) heater control system                             |
| Atlas Copco                            | High-performance BOG compressors for LNG value chains                 | BOG compressor control engineering · global support                  |
| SB Sunbo                               | BOG compressors for LNG carriers                                      | Design, supply and commissioning of BOG compressor control           |

※ Gas supply and safety control on LNG-fuelled vessels is the foundation of our hydrogen facility control.

# From hydrogen stations to overseas plants, energy and industrial references



## SK E&S liquid hydrogen refueling station

VFD, PLC and UPS control system · in commercial operation



## BOBST Korea packaging machinery

Engineering and supply of specialized machine control (servo/motion)



## KOSPO fuel cell power plant CCUS BOP control

BOP control system for carbon capture, utilization and storage · SK ecoplant

## Other major references

- Hyundai Motor Group Metaplant America (USA) — integrated control systems for EV production
- Dongwha formalin resin plant — entire control system designed and built
- KOS · Kiswire — drawing, stranding, plating and heat-treatment lines (Korea, Vietnam, USA, Czechia, Hungary, Malaysia, China)
- KSB — control consoles and system panels for marine application
- Auto doffer, high-density cleaner, thermoplastic spacer, SMR refueling machine — servo/motion control



# Prepare for equipment failure **with data**, not guesswork.

## ISSUE 01

### A machine stops at dawn and no one knows why

Crews arrive only after the failure, and lost production grows with every hour of downtime.

## ISSUE 02

### When experts leave, their know-how leaves too

Judgments made by sound and feel are never recorded.

## ISSUE 03

### Signs of overheating and overload go unrecorded

Without a history of temperature, current and vibration, prevention is hard.



Checking equipment condition with data · illustration

## 01 · BUILD

### Intelligent Control Panel

- New installation or panel replacement
- PLC programming · FAT · commissioning
- Start with 1-2 machines, then expand

## 02 · DATA

### Data service

- Stored in the cloud or on customer servers
- Dashboards · early-warning alarms
- MES · analytics integration (API)

## 03 · OPERATE

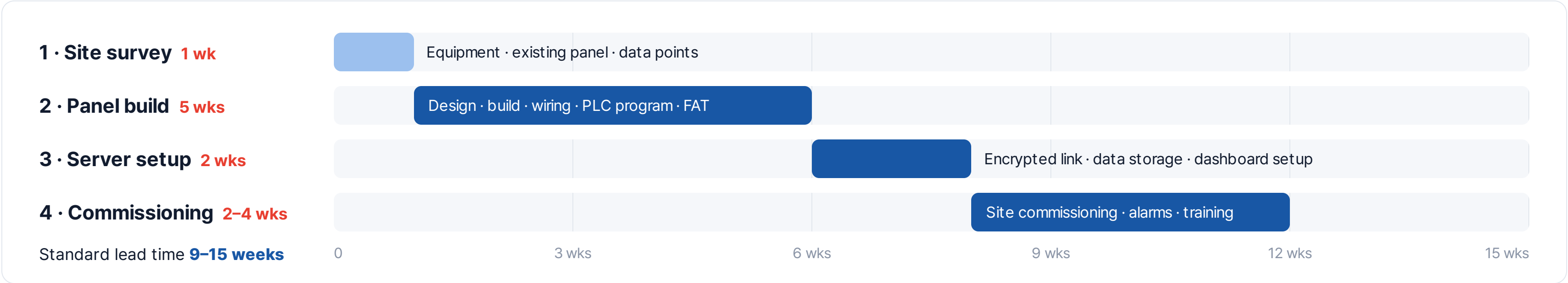
### Maintenance

- Remote status checks · regular inspection
- Alarm tuning · operator training
- On-site support · parts replacement

Cost depends on the equipment and motor ratings, data points, network environment and whether an existing panel is replaced. We provide a configuration and quote after a site survey.

Process

# From site survey to after-sales, one company all the way



May vary with site conditions. Panels are built and tested in our factory before installation, and replacement work is scheduled around your production plan.

**Government support**

Registered smart factory supplier (Korea)

**1 Check programs**

We review available smart factory support programs with you

**2 Proposal · quote**

Configuration tailored to your project plan

**3 Application**

Support with the technical documents required

**4 Build**

Build · commissioning · after-sales once selected

**Roadmap**

Intelligent Control  
Panel · data · AI

**2026 · NOW**

**Data infrastructure**

PLC → DataHub → dashboard, AI anomaly detection  
PoC

**2027**

**Predictive maintenance**

Anomaly detection on real data, digital twin integration

**NEXT**

**Smart factory expansion**

Integrated control of field equipment such as cobots

# We handle the OT layer of smart factory and AI factory projects

From control panel design and build to PLC integration and a standard data API, we handle the field side so you can focus on your platform and services.

## PAIN POINTS

### If your projects keep stalling in the field

- ✓ Every site has different PLC models and protocols
- ✓ Installing and maintaining gateways is a burden
- ✓ It is hard to take on electrical safety for panel work
- ✓ Data formats differ by machine and are hard to organize
- ✓ You lack staff for installation and commissioning
- ✓ You have a platform but need manufacturing sites

## DONG-IN ENSIS

### Field data · OT

- Panel design · build · install · commissioning
- PLC · drive · sensor integration
- Encrypted transfer · DataHub (cloud / on-prem)
- Standard data model + API

## IT PARTNER

### Platform · services

- MES · ERP · EAM integration
- AI analytics · predictive models
- Digital twin · dashboards
- Customer-facing platform operation

## MODEL 01

### OT delivery

We handle control panels, PLCs and on-site installation in your project.

- Panel design · build · install
- PLC · sensors
- Commissioning · O&M

## MODEL 02

### Data integration

Equipment data collected by the panel is delivered to your platform through a standard API.

- Standard data model
- REST API · WebSocket
- MES · AI · digital twin

## MODEL 03

### Joint proposals · sales

We find manufacturing customers and government programs together and propose jointly.

- Joint customer discovery
- Support programs
- Demo equipment

# Different equipment, the same data structure

- 01

**Grouped by equipment**  
site\_id and equipment\_id identify which machine at which site.
- 02

**One format for every sensor**  
Temperature, current or vibration — every sensor uses the same value, unit and status format.
- 03

**Status from thresholds**  
Each sensor has warning and alarm thresholds; if any sensor warns, the equipment status becomes warning.

```
// Example data (for illustration)
{
  "site_id": "PLANT-01",
  "equipment_id": "PUMP-01",
  "timestamp": "2026-10-07T09:30:00+09:00",
  "status": "warning",
  "sensors": [
    { "id": "bearing_temp", "value": 48.2,
      "unit": "°C", "status": "normal" },
    { "id": "motor_current", "value": 31.6,
      "unit": "A", "status": "normal" },
    { "id": "vibration", "value": 4.8,
      "unit": "mm/s", "status": "warning" }
  ]
}
```

**REST API**  
Equipment and sensor lists, latest values, history

**WebSocket**  
Real-time values and status changes

**CSV**  
Export for testing and analysis

**How we work**  
From inquiry to joint projects

1 Inquiry

Share your solution and target project

2 Tech meeting

Agree scope and roles (NDA if needed)

3 Demo · test

Verify integration with demo equipment and test data

4 Joint project

Customer deployment, joint proposal and delivery

**Resources**  
donginensis.com  
IT partner page

PDF

**Data integration overview**  
Architecture · data model · security

JSON

**Sample data (real-time)**  
1 machine · 3 sensors · thresholds

CSV

**Sample data (time series)**  
24 h · 10-min interval · warning

SEC

**Security overview**  
Network · encryption · access

DONG-IN ENSIS · Company Profile

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connecting data.**



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